

Design Goals in Software Engineering

Definition

Design Goals are the objectives that software designers aim to achieve while creating a software system. They help ensure the software is efficient, reliable, and easy to maintain.

Main Design Goals

1. Correctness

- The software should perform the required functions accurately.
- It should meet all user requirements.

2. Reliability

- The software should work consistently without failures.
- It should produce correct results every time.

3. Efficiency

- The software should use minimum resources such as memory, CPU, and time.
- It should provide fast performance.

4. Maintainability

- The software should be easy to modify, update, and fix.
- Future changes should be simple to implement.

5. Reusability

- Software components should be reusable in other projects.

- Reduces development time and cost.

6. Flexibility

- The software should easily adapt to new requirements and changes.

7. Simplicity

- The design should be easy to understand and implement.
- Avoid unnecessary complexity.

8. Security

- The software should protect data and prevent unauthorized access.

Next QUESTION

Cause-Effect Graph (CEG)

Definition

A **Cause-Effect Graph** is a software testing technique used in **Black Box Testing**. It shows the relationship between **inputs (causes)** and **outputs (effects)** using a graph.

- **Cause** = Input condition
- **Effect** = Output or action produced by the system

The graph helps testers create effective test cases.

Why is it Used?

- To identify test cases systematically.
 - To represent complex logical conditions.
 - To ensure all input combinations are tested.
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Example

Requirement:

A student gets a certificate if:

- He passes the exam, and
- His attendance is above 75%.

Causes:

- C1 = Student passed exam
- C2 = Attendance > 75%

Effect:

- E1 = Certificate issued

Graph Logic:

C1 ----\
 AND ----> E1
C2 ----/

This means the certificate is issued only when both conditions are true.